UAV visual localization

Experiment report · ROS 1 Noetic · Individual derivations and team implementation

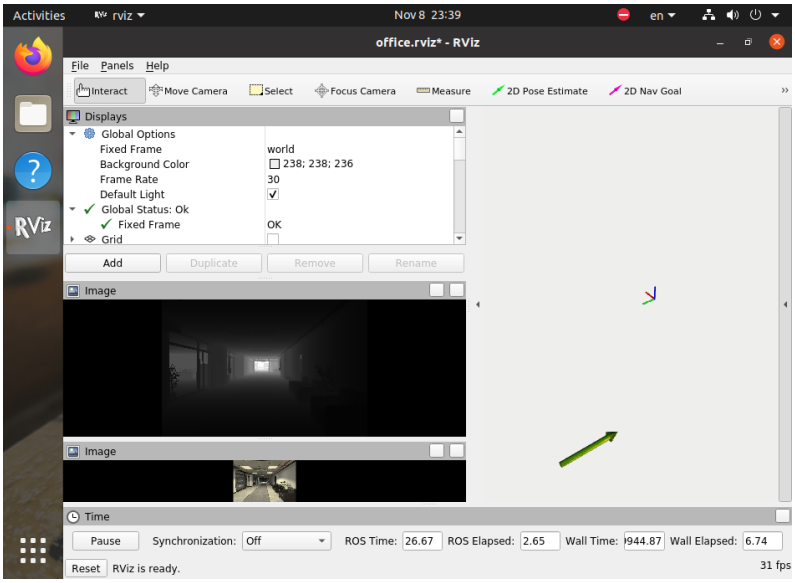
Author: Boyu Huang

The experiment estimates how the camera rotates and moves between consecutive frames. It reuses Lab5 SIFT matches, solves pose with OpenGV, and evaluates it against simulator ground truth.

Data scope: a contiguous prefix of the MIT Office ROS 2 archive, converted to ROS 1. Evaluation covers 2214 pairs up to 135.40 s from the first image. Full-sequence evaluation is pending.

Deliverable	Points	Coverage
D1 Five-point solver	10	Steps, degeneracy, and RANSAC count
D2 Minimal solver	10	Known-rotation derivation and RANSAC
D3 Setup	10	ROS, topics, intrinsics, and depth
D4 2D–2D	40	5pt, 8pt, 2pt, plus no-RANSAC baseline
D5 3D–3D	30	Arun + RANSAC and metric error

Points follow the local course rubric. For team inspection, the demo replays measured RGB, depth and estimated poses.



Experiment preview (Fig. 01). RViz shows Office depth and RGB on the left and camera poses on the right.

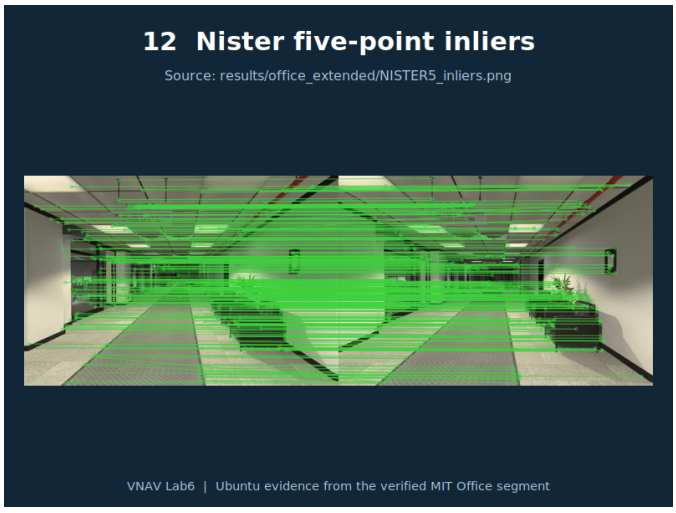
Experiment goals and reading guide

From image correspondences to verifiable camera poses

The goal is to recover camera motion between adjacent images. SIFT proposes which pixels represent the same scene point, then the pose solver uses those correspondences to estimate the motion.

Stage	Input	Output
Match	Two RGB frames	Pixel pairs and descriptor distances
Calibrate	Pixel locations and K, D	Normalized camera bearings
Solve	Bearings, or depth-backed 3D pairs	Relative rotation and translation direction or metric translation
Evaluate	Estimate and odometry reference	Per-pair RPE, solver status and timing

Content	Pages
Tasks, data flow, coordinate convention and algorithm comparison	1–4
D1 and D2 derivations and sampling analysis	5–7
Setup, input checks and implementation	8–10
2D and 3D results, error distributions and comparison	11–14
Reproduction, RViz and match evidence	15–20
Verification and references	21



Matching example (Fig. 12). Green lines connect matches retained by five-point RANSAC.

Coordinates, scale and reference

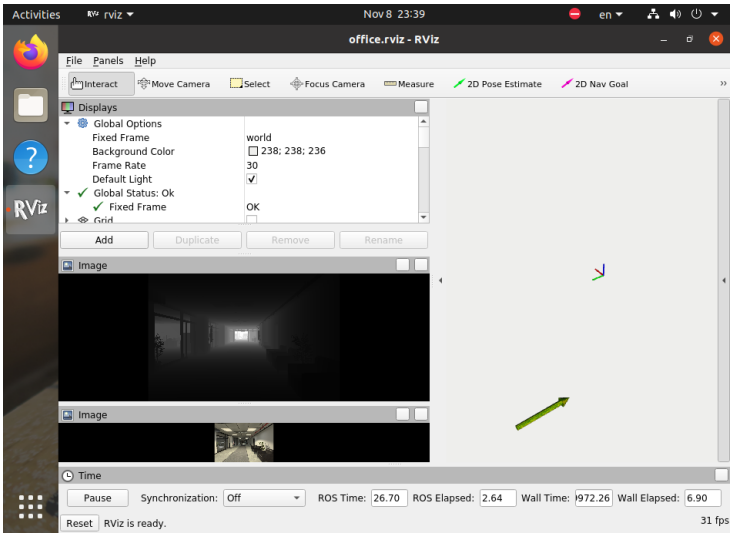
Fix the transform direction before comparing errors

Let T_ab map coordinates in frame b into frame a. Odometry supplies T_world_body. Composing it with the fixed T_body_camera gives the camera pose in the world. The relative reference is inv(T_world_prev) · T_world_curr.

If the previous camera is at the origin and the current camera moves 0.10 m along its x-axis, the relative translation is (+0.10, 0, 0) m. Applying the transform maps a current-camera point into previous-camera coordinates.

Quantity	Computation	Interpretation
Rotation RPE	Frobenius distance of two rotation matrices	Zero means identical rotations
Rotation angle	Angle of R_ref^T R_est	Angular discrepancy in degrees
2D translation direction	Euclidean distance between unit directions	Ranges from 0 to 2, with 2 for opposite directions
3D translation	Compare the two translation vectors directly	Error is measured in meters

Errors are computed first. RViz then anchors each estimate to the previous reference pose and scales the 2D translation direction by the reference length for single-step visualization.



Metric pose example (Fig. 04). Arun estimates translation from depth. RViz anchors the result to the preceding reference pose.

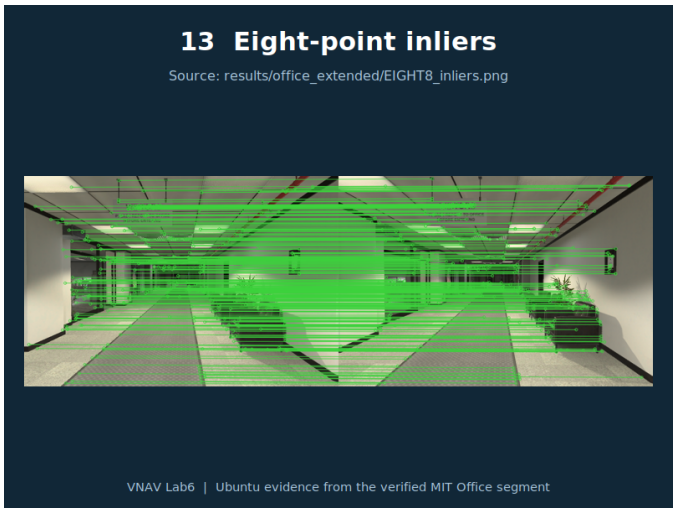
How the four solvers compare

Shared SIFT matches, different motion constraints

Method	Inputs	Output	Main limitation
Nistér 5pt	5 calibrated bearing pairs	Rotation and translation direction	Multiple candidates require disambiguation
8pt	8 calibrated bearing pairs	Rotation and translation direction	Linear system is sensitive to point layout and noise
Known-R 2pt	2 bearing pairs and known rotation	Translation direction	Rotation input error affects translation
Arun 3pt	3 non-collinear 3D pairs	Rotation and metric translation	Requires valid, registered depth

The three 2D methods share the same matches surviving the SIFT ratio test. Arun also requires valid depth on both sides. All methods share the same frame pairs and reference alignment.

The baseline takes one group of points ranked by descriptor distance and proposes a single hypothesis. It shares the five-point geometry and illustrates the effect of repeated sampling and consensus scoring.



Eight-point output (Fig. 13). The model retains inliers from SIFT matches for this pair.

D1 · Nistér's five-point solver

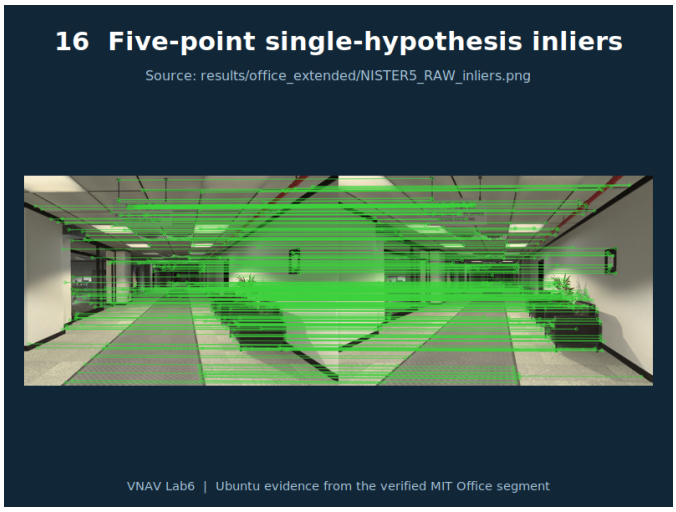
Five correspondences constrain a relative pose

Use R and t to map camera 2 into camera 1. For calibrated unit bearings x1 and x2, x1^T E x2 = 0 with E = [t]_cross R. Rotation has three degrees of freedom and translation direction has two.

Stage	Computation
1. Null space	Stack the epipolar constraints into a 5×9 matrix. With rank 5, express E using four null-space basis matrices.
2. Essential constraints	Fix the global scale and impose det(E)=0 and 2EE^TE-tr(EE^T)E=0. Eliminate variables to obtain a degree-ten polynomial.
3. Physical pose	Back-substitute real roots. Decompose E into candidate rotations and ±t, check positive triangulated depth, then score on further correspondences.

Degeneracy: pure rotation aligns rays after R and leaves translation direction unobservable. Nearly collinear points or very small parallax make estimates unstable. A plane can yield finite solutions, sometimes with multiple compatible candidates. [2]

For independent five-point draws at inlier fraction w, the success probability is about w^5 and the expected wait is 1/w^5. At confidence p, N = ceil(log(1-p) / log(1-w^5)). For w=0.7 and p=0.99, the mean is 5.95 and the budget is 26. The OpenGV NISTER wrapper adds three pairs for decomposition and disambiguation, so it draws eight.



Single-hypothesis five-point baseline (Fig. 16). Green lines mark matches explained by that hypothesis.

D2 · Two points with known rotation

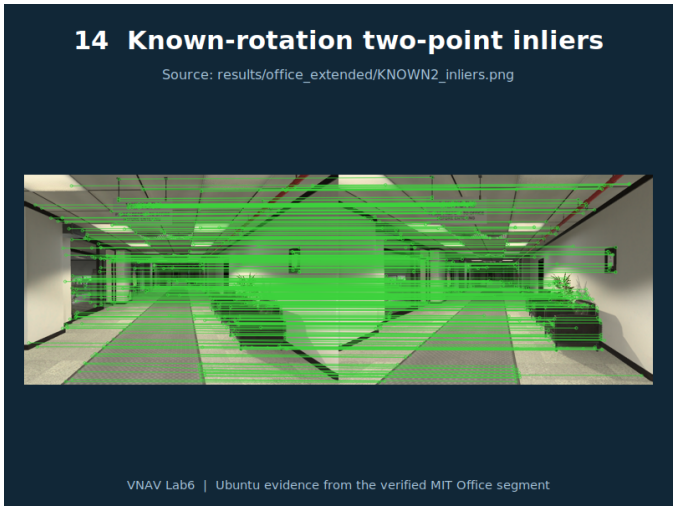
Two constraints determine the direction of translation

With R supplied by the IMU, set n_i = (Rx2_i) × x1_i. Each ray pair gives n_i^Tt = 0, a plane containing translation. The two planes intersect along t = ±(n1 × n2) / ||n1 × n2||.

The normals must be independent. Reject zero-parallax or parallel-normal samples. Use cheirality to select the sign. A refinement can take the right singular vector of the inlier-normal matrix with the smallest singular value.

RANSAC	Procedure
Sample	Draw two matches, reject degenerate normals, form a direction.
Score	Triangulate and use bearing reprojection cost; count residuals below the threshold.
Keep	Keep the largest consensus set; use residuals to break ties if needed. Update the budget with w^2.
Return	Return the best direction and inliers, optionally refit using all inliers.

The two-point experiment uses ground-truth relative rotation as the known IMU rotation. Its near-zero rotation error follows from this input.



Two-point output (Fig. 14). With rotation supplied, the matches constrain the translation direction.

Sampling budget and outliers

Cleaner matches make an all-inlier draw more likely

For inlier fraction w and sample size m, independent draws give an all-inlier probability of w^m. All N attempts fail with probability (1–w^m)^N. Requiring this to be at most 0.01 yields a 99% confidence budget.

Inlier fraction	Two points m=2	Five points m=5	Eight points m=8
50%	17	146	1177
70%	7	26	78
90%	3	6	9

These are theoretical budgets. OpenGV's central five-point and eight-point wrappers draw additional matches for decomposition or candidate selection. The experiment caps iterations at 300 and uses inlier consensus to estimate when to stop.

Several mismatches can still fit an approximate geometric model. Assess inlier counts together with rotation and translation errors and scene parallax.

D3 · Build and camera calibration

Build outputs and run logs provide setup evidence



Fig. 09 Catkin build output displayed on the Ubuntu desktop.

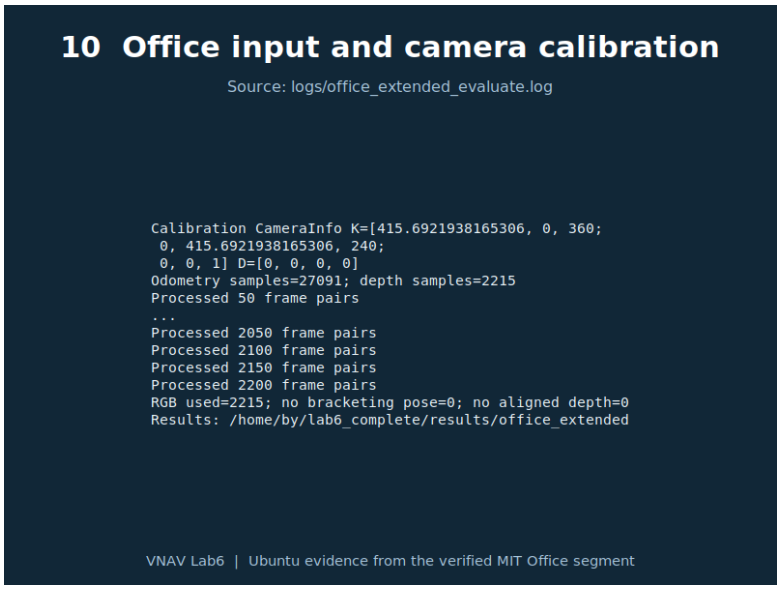


Fig. 10 Actual evaluation log showing CameraInfo, depth samples and frame counts.

D3 · Data topics and alignment

Check whether pose and depth can be aligned to RGB

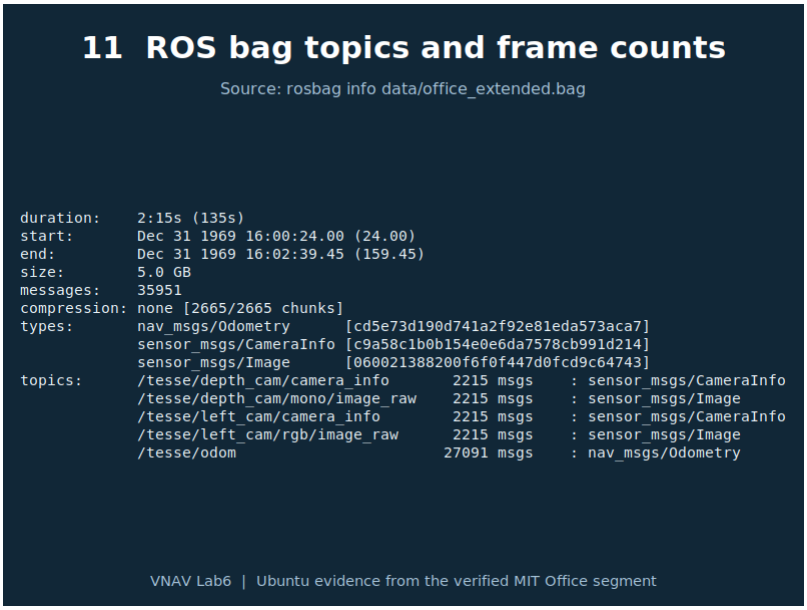


Fig. 11 rosbag info shows input topics and counts for the Office segment.

Alignment step	Check
Adjacent RGB frames	Process in time order and skip duplicate or backwards timestamps
Associate depth	Choose nearest depth within 15 ms and require valid depth at both pixels
Associate reference	Use bracketing odometry and interpolate translation and quaternion rotation

D3–D5 · Data and implementation

From pixels to relative transforms

Stage	Implementation and parameters
Input	Ubuntu 20.04, ROS Noetic, OpenCV 4.2 contrib, OpenGV. Read left RGB, registered depth, CameraInfo and ground-truth odometry.
Calibration	fx=fy=415.6922, cx=360, cy=240, zero distortion. Undistort and normalize (x,y,1) to unit bearings.
Alignment	Match depth timestamps within 15 ms. Interpolate odometry translation linearly and rotation by SLERP. Apply the starter camera-to-body extrinsic.
2D–2D	Reuse Lab5 SIFT, FLANN and the 0.8 ratio test. OpenGV Nistér, eight-point and known-rotation two-point use RANSAC, threshold 5e-6 in the two-view cost.
3D–3D	Back-project with Z depth and reject invalid samples. Arun aligns centered 3D points by SVD, corrects reflections and recovers metric translation. RANSAC threshold 0.04 m.
RPE and RViz	Compare previous-to-current relative transforms. Rotation uses Frobenius distance; 2D translation uses unit directions and 3D uses meters. RViz is anchored to the previous ground-truth pose; 2D display also uses its scale.

Skip 2D direction error below a 1 mm ground-truth baseline. Failures remain in the CSV as gaps. RANSAC uses up to 300 iterations at 0.99 confidence. The baseline scores one hypothesis from the lowest descriptor-distance matches.

D4 · 2D correspondence results

Three RANSAC methods and a single-hypothesis baseline

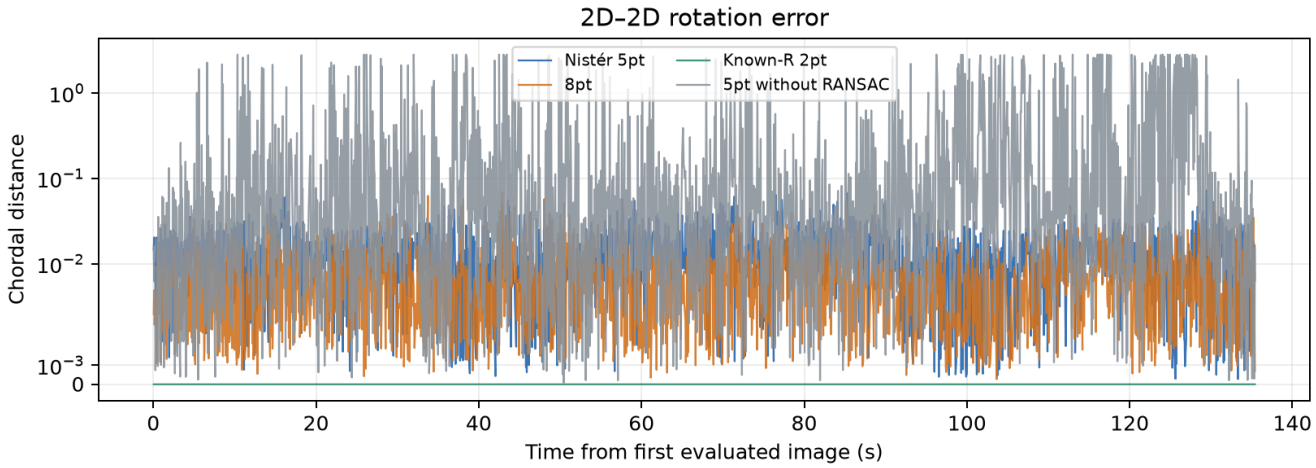


Fig. 05 Chordal rotation error for the 2D methods.

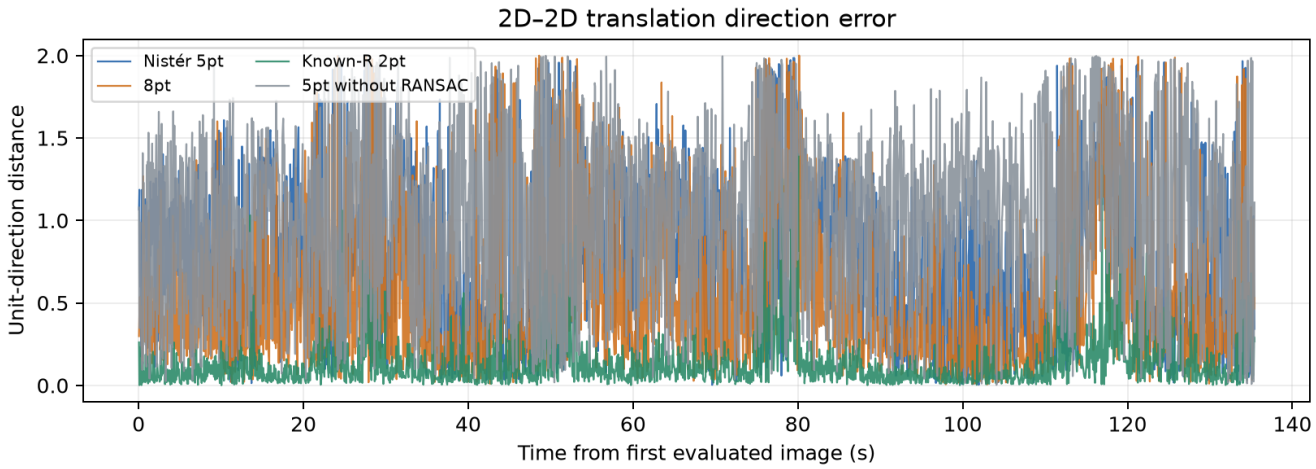


Fig. 06 Translation direction error, ranging from 0 to 2.

Method	Solved	Dir. samples	Median rot. °	Median dir.
Nistér 5pt	2214/2214	2214	0.4978	0.7523
8pt	2214/2214	2214	0.2309	0.4252
Known-R 2pt	2214/2214	2214	0	0.08999
5pt without RANSAC	2211/2214	2211	1.391	1.084

With low parallax, small pixel errors can produce large direction errors. Interpret direction and metric errors in their respective units.

D5 · 3D correspondence results

Arun + RANSAC recovers metric translation

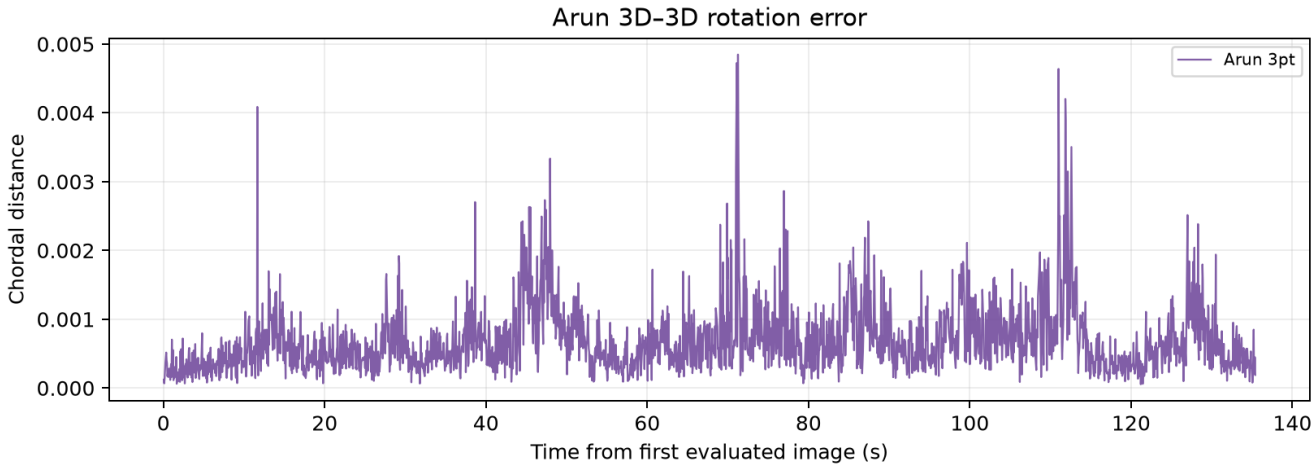


Fig. 07 Arun chordal rotation error.

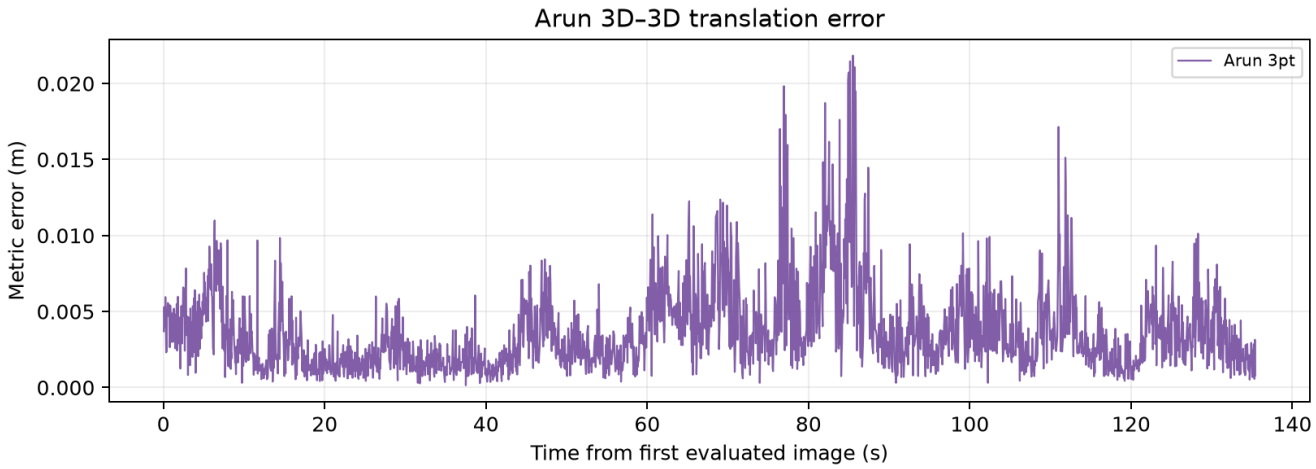


Fig. 08 Arun metric relative translation error.

Solved	Median rot. °	Median trans. m	Median inliers	Median solve ms
2214/2214	0.02265	0.002914	246	0.03213

Depth turns pixels into points with a metric scale. Metric transforms can be composed across frames, but drift accumulates. The RViz display shows ground-truth-anchored single-step poses.

Error distribution and target rates

Medians describe typical errors; the 90th percentile describes the upper tail

Method	Median °	P90 °	Below 1°
Nistér 5pt	0.4978	1.097	1913/2214 (86.4%)
8pt	0.2309	0.6933	2129/2214 (96.2%)
Known-R 2pt	0	0	2214/2214 (100.0%)
Arun 3pt	0.02265	0.05165	2214/2214 (100.0%)
5pt without RANSAC	1.391	39.49	952/2211 (43.1%)

2D method	Median	P90	Below 0.5
Nistér 5pt	0.7523	1.375	762/2214 (34.4%)
8pt	0.4252	1.323	1254/2214 (56.6%)
Known-R 2pt	0.08999	0.3105	2111/2214 (95.3%)
5pt without RANSAC	1.084	1.69	456/2211 (20.6%)

Arun's median metric error is 0.002914 m, with P90 of 0.007077 m and 2140/2214 (96.7%) of pairs below 1 cm.

Target rates use pairs with finite errors. Solve counts are shown on the preceding pages. The known-rotation solver receives rotation as input. The thresholds here describe the measurements; course acceptance follows the handout.

RANSAC comparison and findings

Compare one hypothesis against repeated sampling with the same five-point geometry

Five-point method	Solved	Median rot. °	Median direction	Median inliers
With RANSAC	2214/2214	0.4978	0.7523	289
Single hypothesis	2211/2214	1.391	1.084	194.5

On this segment, repeated sampling increased median inlier count and reduced both rotation and direction error. The eight-point medians are below the five-point RANSAC medians here. Changes in scene structure or frame-to-frame motion warrant another comparison.

Known rotation reduces the unknowns in the two-point solver. Valid depth gives Arun an actual displacement length, so its errors can be read in centimeters or millimeters. Interpret 2D direction and 3D metric errors separately.

Symptom	What to check first
Large direction error	Check the reference baseline, then whether matches cluster on a single wall.
Few usable Arun points	Check depth validity and registration to RGB.
Pose arrows and curves look different	RViz uses the preceding reference pose as an anchor. The curves compare relative transforms.

Data provenance and live reproduction

Trace inputs, commands and outputs

The evaluation covers 2214 adjacent pairs over about 135.40 s. The main tables and Figs. 05–08 use this run. Figs. 01–04 show RViz playback of a short segment from the same sequence.

Step	Action	Verification
1. Prepare	Use the built Ubuntu ROS Noetic environment	The lab_6 package and Office input bag are available
2. Demo	In ~/lab6_complete, run bash demo.sh ARUN3	RViz displays RGB, depth and estimated poses
3. Recompute	Run bash evaluate.sh	A new result directory contains per-pair CSV and pose bags
4. Verify	Open the summary CSV and input SHA256	Check method names, denominators and the input file

The full archive is about 8.57 GB. This run extracts complete messages from about 5.00 GB of contiguous downloaded bytes. Once available, convert and evaluate the full recording, then regenerate the plots from that run.

RViz replay screenshots

Depth and RGB are on the left; estimated and reference poses are in the center.

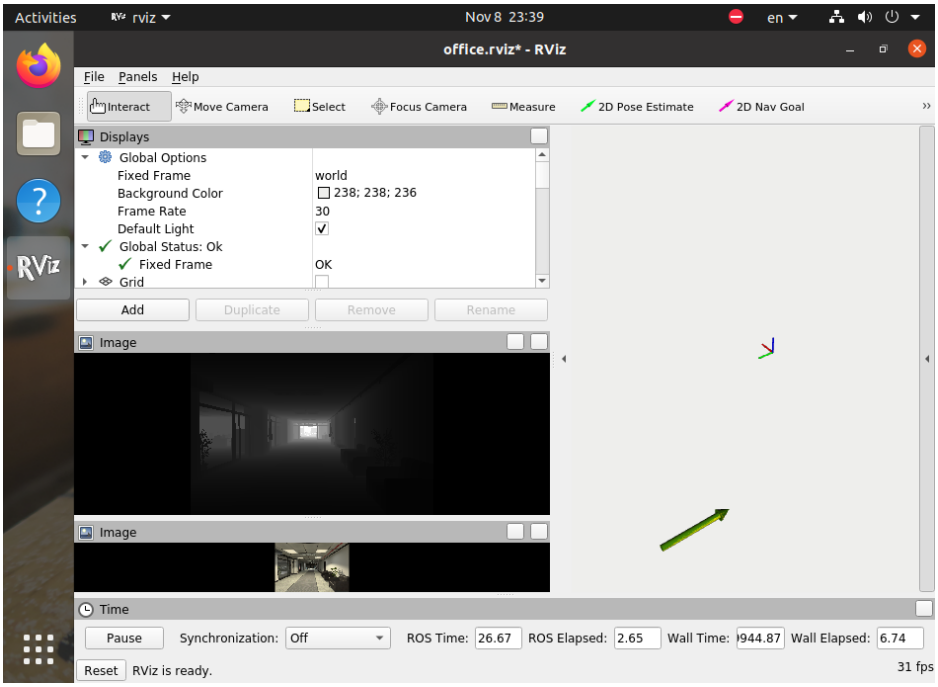


Fig. 01 Nistér 5pt replay of the Office prefix.

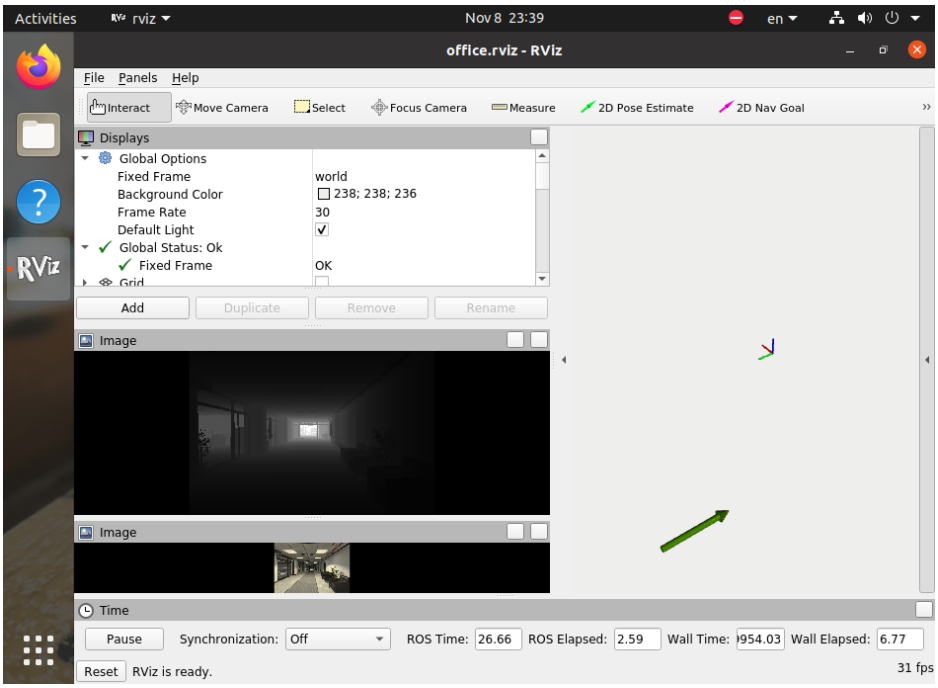


Fig. 02 8pt replay of the Office prefix.

RViz replay screenshots

Depth and RGB are on the left; estimated and reference poses are in the center.

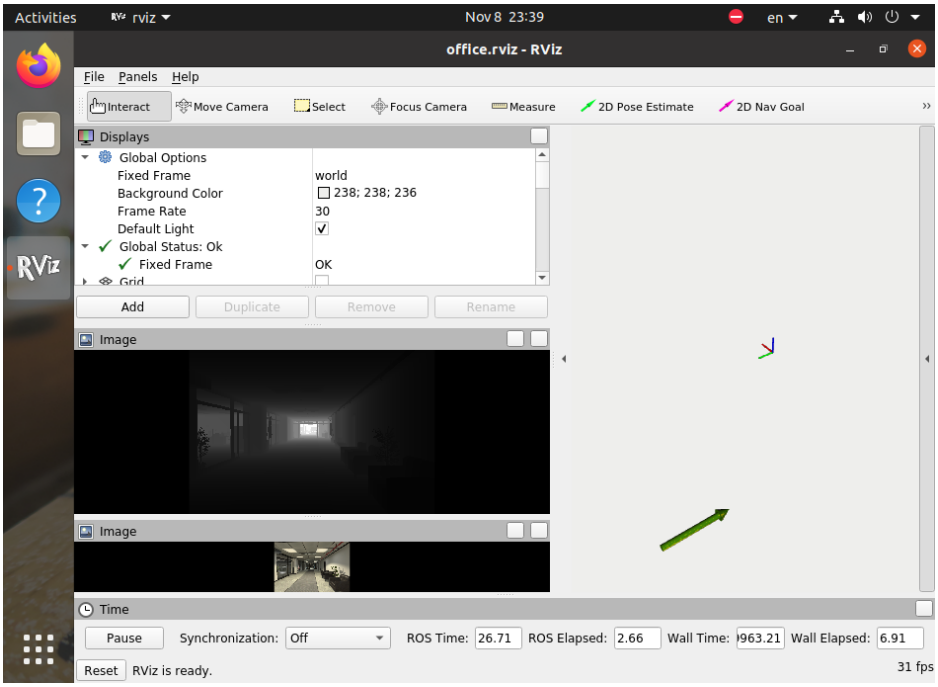


Fig. 03 Known-R 2pt replay of the Office prefix.

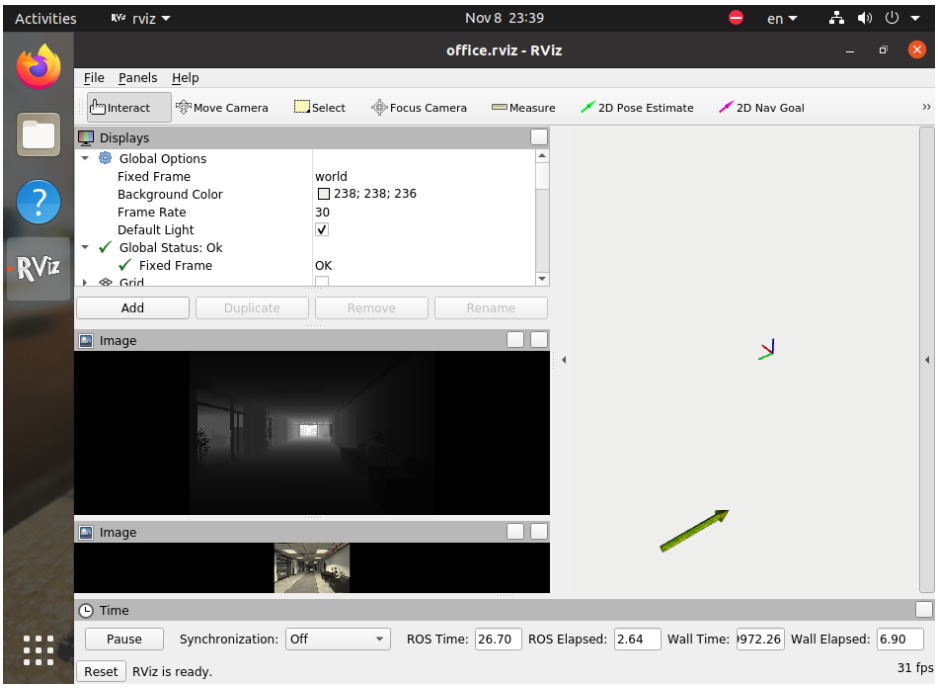


Fig. 04 Arun 3pt replay of the Office prefix.

Match and inlier evidence

Actual Ubuntu output. Green lines connect correspondences retained by each model.



Fig. 12 Representative pair for NISTER5.



Fig. 13 Representative pair for EIGHT8.

Match and inlier evidence

Actual Ubuntu output. Green lines connect correspondences retained by each model.

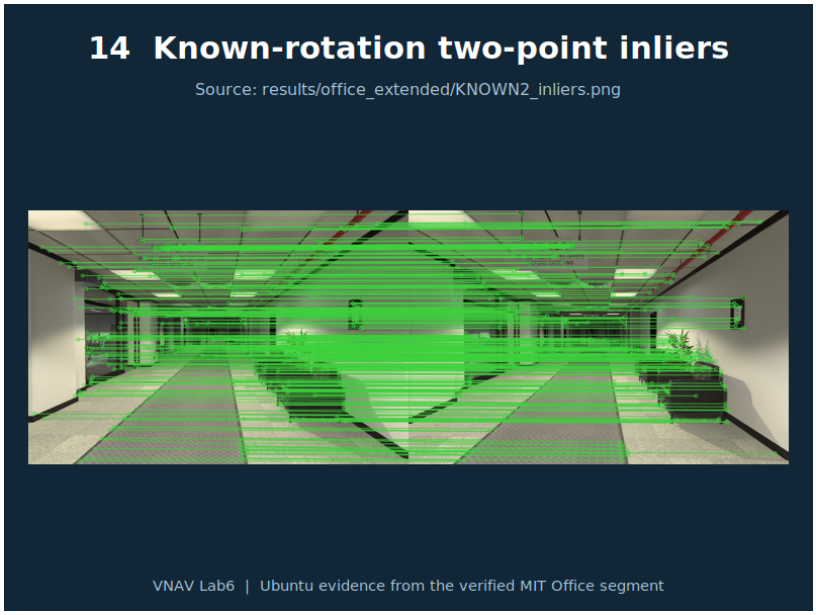


Fig. 14 Representative pair for KNOWN2.



Fig. 15 Representative pair for ARUN3.

Match and inlier evidence

Actual Ubuntu output. Green lines connect correspondences retained by each model.

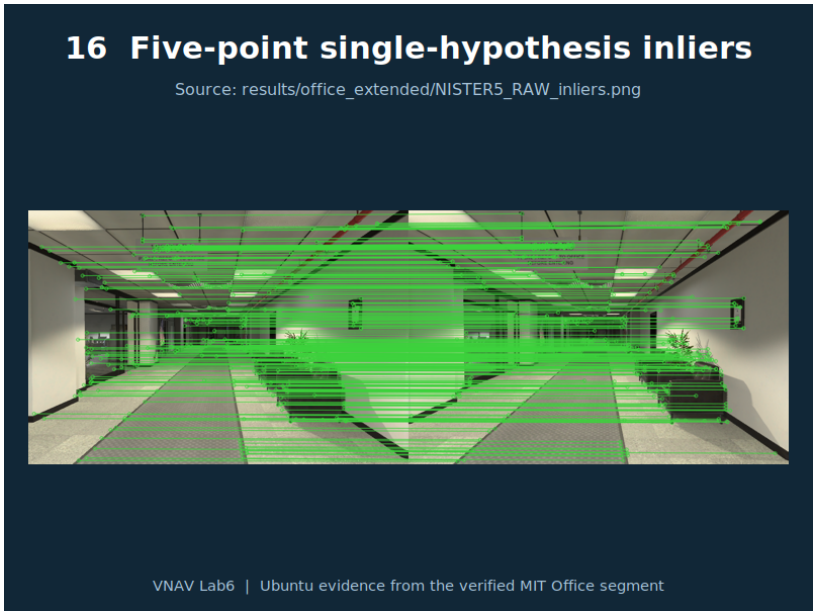


Fig. 16 Representative pair for NISTER5_RAW.

The baseline proposes one hypothesis from the highest-ranked matches. The screenshot shows one successful pair. The error curves and summary cover all evaluated frame pairs.

Check	What to inspect
Matches	Check whether green lines connect corresponding edges or texture
Robustness	Compare with five-point RANSAC direction errors and inlier counts

Verification and references

Reproduction, metrics and limitations

Artifact	Purpose
rpe.csv	Per-pair errors, match counts, inliers, baseline and solver timing.
summary.csv	Valid counts and medians. Timing covers pose solving and RANSAC.
Demo guide	Run RViz, switch methods, recompute and inspect output.

Scope: the original Google Drive file returned 404. The course now provides a ROS 2 archive. The download quota limited evaluation to complete records in the downloaded prefix. The RViz replay screenshots show a short segment of the same sequence. Figures 09–16 show logs and matches from the extended run. An optional Lab4 RGB-D recording failed during depth camera metadata parsing in the older simulator; its log is retained.

Median 2D direction errors were 0.7523 (5pt), 0.4252 (8pt), 0.08999 (known rotation) and 1.084 (no RANSAC). Arun's median metric error was 0.002914 m. The translation metrics use different units.

[1] MIT VNAV, Lab 6 Exercises (ROS 1).

https://vnav.mit.edu/labs_2023/lab6/exercises.html

[2] D. Nistér, An Efficient Solution to the Five-Point Relative Pose Problem, CVPR 2003.

https://www-users.cse.umn.edu/~hspark/CSci5980/nister.pdf

[3] OpenGV, library and sample-consensus implementation.

https://github.com/ethz-asl/opengv

[4] MIT VNAV, Office dataset (ROS 2 archive).

https://drive.google.com/file/d/1TohOnfjR_FKf_y9wYTEkmv-XCuFPFRPt/view

[5] MIT-SPARK, VNAV Lab starter code.

https://github.com/MIT-SPARK/VNAV-labs